

Khoa Nguyen

Solution Architect / Engineering Manager

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Experienced Solution Architect with extensive exposure in designing, developing, and implementing cutting-edge software solutions. Proven ability to guide and mentor development teams, manage all aspects of the software development life cycle, and deliver high-quality results. Expertise in Web technologies, Java, Magnolia DXP and other cloud solutions (Adobe, AWS, Google).

Work Experience

mis.studio

Aug 2021 – Present

Solution Architect

- **Hybrid Cloud & Migration Pattern:** Overcame legacy PaaS rigidity by migrating enterprise Magnolia CMS to a hybrid on-premise cloud, improving data governance and compliance.
- **Multi-Tenant / Global Rollout Pattern:** Solved multi-region fragmentation by consolidating 3+ web estates onto Magnolia PaaS with localized content and in-region hosting.
- **Composable Commerce & Embedded Finance Pattern:** Addressed disconnected buying journeys by integrating Commerce Layer with bespoke Order Management System, enabling embedded financing across 1M+ products.
- **AI & Content Orchestration Pattern:** Streamlined manual editorial workflows by integrating ChatGPT generative AI components and Salesforce Marketing Cloud into CI/CD publishing pipelines.
- **Real-Time Edge & Signage Pattern:** Resolved fragmented physical marketing displays by engineering a centralized digital signage CMS supporting real-time scheduling and dynamic rendering.

POPS Worldwide

Jul 2019 – Jul 2021

Tech Lead

- **High-Concurrency Event-Driven Pattern:** Resolved live traffic spikes of up to 100K concurrent users by building an event-driven system with GCP Pub/Sub, Redis caching, and auto-scaling Kubernetes clusters.
- **Polyglot OTT Services Pattern:** Eliminated monolithic streaming constraints by designing a GCP microservices platform powering video streaming, live broadcasts, and digital comics.
- **End-to-End Content Orchestration Pattern:** Automated manual media workflows by engineering a custom CMS to manage transcoding, scheduling, and multi-device content distribution lifecycle.

FPT Software

Jan 2018 – Jul 2019

Solution Architect / Tech Lead

- **Legacy Cloud Migration & Modernization Pattern:** Resolved legacy technical debt and system silos by migrating monolith CMS and loyalty applications to a cloud-native AWS/Adobe platform for a major Singapore real-estate group.
- **Architecture Estimation & Governance Pattern:** Eliminated project scope slippage by leading technical requirement analysis and establishing Work Breakdown Structures (WBS) for multi-system digital transformations.
- **Distributed offshore Engineering Pattern:** Streamlined cross-border delivery bottlenecks by collaborating with offshore development teams to standardize reusable design templates, components, and service layers.
- **Behavioral Personalization & Analytics Pattern:** Fixed fragmented customer insights by integrating Adobe Target and Adobe Analytics to enable real-time visitor tracking, traffic conversion, and targeted content personalization.

Aleph Labs

Jan 2017 – Jan 2018

Tech Lead

- **High-Security Banking Core Pattern:** Overcame strict enterprise banking compliance requirements by leading cross-functional teams to architect and deliver scalable Adobe Experience Manager (AEM) solutions.
- **Component-Driven Modular Architecture Pattern:** Eliminated monolithic design friction by authoring modular component specifications, design templates, and decoupled service-layer technical blueprints.
- **Closed-Loop Personalization & Analytics Pattern:** Fixed disconnected user tracking by orchestrating Adobe Marketing Cloud (DTM, Target, Analytics) integrations to drive real-time behavioral personalization and conversion insights.
- **Automated DevSecOps & Quality Assurance Pattern:** Reduced manual review bottlenecks and security risks by deploying automated static analysis tools to track test coverage, detect vulnerabilities, and streamline code reviews.
- **Executive Quality Control & Telemetry Pattern:** Resolved operational visibility gaps by engineering real-time analytics dashboards that aggregated code health and security metrics for executive reporting.

Navigos Group

Aug 2014 – Dec 2016

Senior Software Engineer

- **Enterprise Search & Relevancy Pattern:** Resolved inaccurate candidate-job matching by integrating Elasticsearch to optimize job search scoring, query latency, and result relevancy.
- **Automated Unstructured Data Pipeline Pattern:** Eliminated manual data entry overhead by building a resume parsing engine to automatically extract and structure candidate contact information.
- **Big Data & Analytics Processing Pattern:** Resolved fragmented hiring insights by architecting a career analytics platform using Spring Boot and Spring Data to aggregate, store, and analyze job posting datasets.
- **Decoupled Full-Stack Product Pattern:** Streamlined interactive campaign delivery by building a challenge management product leveraging AngularJS and Spring MVC decoupled service layers.

PYCOGROUP

Aug 2011 – Jul 2014

Senior Java Engineer

- **Headless DXP Architecture Pattern:** Resolved presentation-backend tight coupling by building Java Spring backend microservices integrated with an Adobe Experience Manager (AEM) frontend presentation layer.
- **Designed and proposed technical solutions** based on user requirements or specifications. Translated user stories into implementation approaches.
- Defined and unified development tools, workflow, source control, and conventions for the team at the beginning.

Education

HCMC University of Science

2007 – 2011

Bachelor of Information Technology

Languages

English Professional Fluency